



Advanced glaucoma disease segmentation and classification with grey wolf optimized U-Net++ and capsule networks

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Abstract

Purpose To develop an advanced automated glaucoma diagnostic system by integrating an optimized U-Net++ segmentation model with a Capsule Network (CapsNet) classifier, enhanced through Grey Wolf Optimization Algorithm (GWOA), for precise segmentation of optic disc and cup regions and accurate glaucoma classification from retinal fundus images.

Methods This study proposes a two-phase computer-assisted diagnosis (CAD) framework. In the segmentation phase, an enhanced U-Net++ model, optimized by GWOA, is employed to accurately delineate the optic disc and cup regions in fundus images. The optimization dynamically tunes hyperparameters based on grey wolf hunting behavior for improved segmentation precision. In the classification

phase, a CapsNet architecture is used to maintain spatial hierarchies and effectively classify images as glaucomatous or normal based on segmented outputs. The performance of the proposed model was validated using the ORIGA retinal fundus image dataset, and evaluated against conventional approaches.

Results The proposed GWOA-UNet++ and CapsNet framework achieved a segmentation and classification accuracy of 95.1%, outperforming existing benchmark models such as MTA-CS, ResFPN-Net, DAGCN, MRSNet and AGCT. The model demonstrated robustness against image irregularities, including variations in optic disc size and fundus image quality, and showed superior performance across accuracy, sensitivity, specificity, precision, and F1-score metrics.

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Conclusion The developed automated glaucoma detection system exhibits enhanced diagnostic accuracy, efficiency, and reliability, offering significant potential for early-stage glaucoma detection and clinical decision support. Future work will involve large-scale multi-ethnic dataset validation, integration with clinical workflows, and deployment as a mobile or cloud-based screening tool.

Keywords Automated diagnosis · Retinal image analysis · Medical image segmentation · Vision preservation · Grey wolf techniques · CapsNet

Introduction

Retinal ganglion cell loss causes the incapacitating and painful visual neuropathy known as glaucoma. Untreated, it could cause notable visual handicap. The anatomical changes to define clinical manifestations of glaucoma and should guide doctors in making informed clinical judgments of this neuropathy condition. Before making clinical decisions concerning any instance glaucomatous ocular neuropathies, a thorough evaluation of symptoms must be done to enable informed clinical judgements [1, 2]. Currently, Glaucoma diagnosis calls for numerous tests in addition to compiling a lot of information from eye exams; can sometimes offer a difficult task. It is also challenging to separate early-stage glaucoma characteristics from those observed in healthy people; additionally, in order to track the progression of the disease between tests, attempts researchers are developing artificial intelligence diagnostic systems which can separate between genuine disease indications from regular human body changes [3].

While there is no cure for glaucoma, its symptoms can be significantly reduced with early detection and prompt treatment. Consequently, the development of automated tools for early glaucoma identification is essential [4]. Retinal fundus scans offer crucial data that enables evaluation of many eye components, including the macula, vitreous retina, and nerve blood vessels. Fundus cameras are frequently used by optometrists to acquire images of the retina for use in diagnosing conditions like glaucoma. Improved diagnostic techniques must be developed because glaucoma remains a key factor behind blindness. When it comes to sending images from retinas to brains, optic

nerves are essential. New advances in detection of glaucoma involve multi-objective optimisation and machine learning to improve the diagnostic accuracy and operational performance. In [5] proposed a multi-objective evolutionary algorithm for detection of glaucoma based on learning in retinal fundus image with superior optimization of feature selection and classification accuracy that highlights importance of trade-off balancing during the medical image analysis.

Healthcare professionals utilize the Cup-to-Disc Ratio (CDR) as a key visual test to identify glaucoma conditions. The majority of glaucoma symptoms appear at the Optic Disc (OD). The accuracy of automated techniques that depend on segmenting an OD is crucial; even little deviations can significantly change the diagnosis [6]. Compared to its predecessors, localization techniques are less likely to make mistakes since they offer crucial contextual information about the object's precise placement inside a picture as well as surrounding features [7]. For automated classification, identifying delicate and It is difficult for practitioners to analyze complex retinal fundus images which provide indicators of disease. Natural settings are easily recognized due to their vibrant colors, shapes, and textures, yet medical photos can hide significant disease symptoms that must be interpreted by experts. However, recent developments in Deep Learning have demonstrated that it can provide precise representations that enable the identification of minute variations [8]. Because of this, their depictions are incredibly succinct and educational. In [9] developed a strong framework for automated glaucoma classification using various machine learning and deep learning tools in their subsequent research, highlighting the critical role of architecture design in diagnostic results.

By reducing human error, accelerating health care delivery in underserved areas, relieving medical staff stress, and eliminating bias or discrimination, Computer-Assisted Diagnosis (CAD) devices could significantly improve the efficiency and quality of medical screening programs Modern glaucoma diagnosis depends on the Cup-to-Disc Ratio measurement which shows symptoms directly on the Optic Disc (OD). The precision of automated diagnostic methods heavily relies on accurate segmentation of Optic Disc structures because small errors result in drastic impacts to diagnostic outcomes. Localization

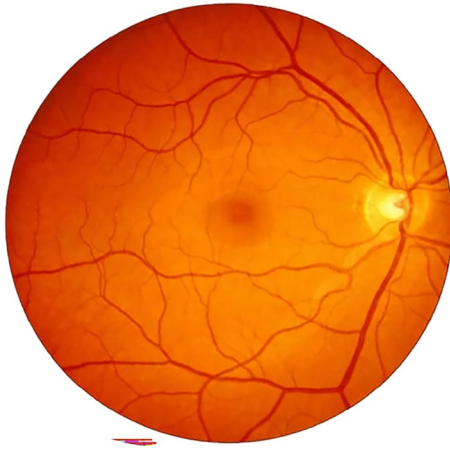


Fig. 1 Retinal fundus image

techniques deliver essential detection information because they identify the precise position of the Optic Disc and show its interconnections with image elements thus enhancing the detection process accuracy. The automatic identification of retinal fundus images proves to be a major difficult task.

Due to subtle indicators that may stay hidden in medical images the process needs advanced expert experience to achieve proper interpretation. Natural scenes from Fig. 1 contain distinct attributes such as color and shape and texture that enhance identification of pathological features. Recent developments in DL have demonstrated their capacity to identify precise data representations that enable the identification of minute differences, making these representations incredibly succinct and instructive. Computer-Assisted Diagnosis (CAD) tools have the potential to significantly increase medical screening programs' effectiveness and caliber. Medical systems using AI and robotics technology provide three main benefits through error reduction and faster care delivery to underserved areas together with decreased professional strain and unbiased medical action [10].

To improve the disc as well as optic cup segments while reducing computing time, deep learning approaches have become more popular recently. This novel method uses Convolutional Neural Networks (CNNs) to segment discs and cups using the RIM-ONE dataset, and the results are amazing. One particularly useful dataset, the RIM-ONE dataset, allowed our model to attain remarkable disc segmentation accuracy with previously unheard-of ease.

The fundamental component of this innovative method is the use of a convolutional neural network based on ensemble learning to segment fundus pictures to identify the glaucoma. There are two essential elements in the multi-step process. Splitting retinal optic disc images comes after determining the ROI in retinal pictures. Using M-Net for implementation is a crucial component of this detection technique. Recently, an extra phase known as Polar Transformation was added to the M-Net methodology to boost its overall efficacy [11]. Prior to starting the M-Net testing and learning phases, the area around it will be extended for transformation using Polar methods. Traditionally, deep learned methods has mostly concentrated on using optical discs to retrieve data. Recent developments, however, may make it possible to broaden this method by using optical cup with disc segment as a tactic to increase accurate when glaucoma is diagnosed in its early stages. As a result, a novel deep learnings technique has developed that applies the tactic to the segment in disc and cup as reference [12] explains. The advanced diagnostic approach improves both diagnosis accuracy and efficiency during early glaucoma detection and fosters progress in the field of study. A novel DL framework for glaucoma optic disc haemorrhage segmentation. The designed framework uses integrated expert knowledge to segment the fundus images. It is used to segment the blood vessel and optic disc which are presented in the given fundus and retinal images. Experimental results show that the designed framework maximizes the accuracy in image segmentation.

Deep learning methods have achieved substantial progress in detecting glaucoma with their current techniques. The application of deep learning supports patients in receiving early vision preservation through the analysis of glaucoma progression by studying retinal fundus images.

- A purpose-built segmentation framework segmented the optic cup region found in retinal fundus images with high efficiency. The use of UNet++ architecture helps the method obtain crucial cup-and-disc features necessary for evaluating glaucoma. The diagnostic quality and detection efficiency of glaucoma decreases because of poor performance in the baseline UNet++ segmentation results. The adaptation of the adaptive GWO algorithm with U-Shaped Network helps

optimize hyperparameters for better segmented accuracy which leads to superior early glaucoma diagnosis results.

- The glaucoma diagnosis field has seen a major breakthrough with the Capsule Network (CapsNet) architecture design. CapsNet maintains retinal feature spatial relationships to deliver an efficient framework that detects and classifies glaucomatous abnormalities. The standard performance metrics accuracy and sensitivity together with specificity and F1-score were used to evaluate the reliability and clinical effectiveness of our CapsNet-based model against traditional glaucoma detection techniques.

The research discoveries have made substantial progress which may revolutionize glaucoma detection at its early stages and major strategies exist today to protect susceptible individuals whose vision faces danger from degenerative diseases.

The study has three main research objectives that form the core focus of its investigation. This study requires three specific questions that will steer our investigation in the right direction. The main objective of our research is to enhance glaucoma segmentation and classification methods. Given three distinct techniques form our investigation methodology to reach our objective.

- (1) The main purpose is to improve both detection reliability and robustness for glaucoma outcome assessments.
- (2) The proposed research combines Grey Wolf Optimization with UNet++ techniques to better segmentation results.
- (3) This research aims to develop two areas of glaucoma detection with Capsule Networks by enhancing segmentation and classification performance.

Literature survey

The designed for semi-supervised graph-based network for optic discs. It is used to address the segment issues which cause damage to the segmentation process. The designed network identifies the boundary regions and characteristics values which provide

relevant data for disc segmentation. An important region of interest (ROI) is also identified which eliminates unwanted energy consumption in the computation process. The designed network enlarges the performance and effectiveness ratio of the segmentation in discs. As demonstrated by [13], computer-aided techniques for intelligent glaucoma diagnosis have become more and more popular in recent years.

A deep learning methodology performs glaucoma disease diagnosis. The diagnosis of glaucoma requires substantial time along with precise data to achieve efficient disease prediction. Optic cup image cropping enables this method to perform blood vessel segmentation for detection tasks. The DL-based approach makes it possible to deliver medical and diagnostic services more feasibly to users. The method improves the precision of determining upcoming diseases. The algorithm unites for image processing techniques with machine learning to achieve its objectives. Unsupervised learning algorithms use retinal images to divide the enhanced images into optic disc and cup elements according to the research [14]. Multiple machine learning techniques determine whether images are glaucoma or normal.

The authors in [13] tackled the long processing time by creating a multi-resolution image fusion technique which uses an attention-based residual convolution module using large kernels. Picture analysis depends on the measurement of the Cup-to-Disc Ratio (CDR) because it functions as the essential factor for segmented images. The module in convolution attention serves to improve medical image high clarity ranges. The utilization of CDR enables this approach to lower its processing time requirements and computational expenses during disease detection procedures. The field of glaucoma diagnosis now utilizes multiple classifiers with Multi-layer Perceptron Random Forest classifiers and Radial Basis Function classifiers among them [15]. The diagnostic techniques for glaucoma identification combine heuristic methods with deep learning technologies. The screening process using heuristic methods allows assessments of retinal nerve fiber layer thickness while extracting features from processed images. Glaucoma diagnosis became more accurate due to the research done in [16] which studied higher-order spectral combined with textural data analysis methods. Fundus pictures segmentation depends on proper detection of Optic Cup (OC) and Optic Disc (OD) according

to [17]. The proposed research employs attention-based U-Net architecture implementing CNN blocks of VGG19, Inception-V3, and ResNet50 for fundus image segmentation purposes. This research used different data augmentation approaches to stop overfitting along with reaching high accurate rates.

In a recent work done by authors [18], the three most commonly used CNN architectures were used: Inception-v3, VGG19, and ResNet50. Their performances were compared and assessed with the view of determining how efficient the CNN-based frameworks can be in detecting glaucoma cases in its automated manner. The approach uses a limited subset of visual attributes thus achieving low classification performance. Joint segmentation of optic disc and cup occurs through the method by analyzing features at different resolution levels. The transition cuts down on the analysis expenses linked to optic disc and cup segmentation through fundus image feature evaluation. This research method extends the functional needs of fundus pictures in segmentation procedures which results in better practical application of segmentation techniques [19].

Techniques based on deep learning provide an additional technique for detecting glaucoma. Research has centered on this type of identification using deep learning models that can automatically partition the surfaces of optic and disc-cups [20]. In [21], the use of a nature-inspired computing technique led to increases in glaucoma classification due to efficient address of feature relevance and redundancy. However, these studies typically just address optic discs as probable glaucoma indications without providing a comprehensive answer. However, a comprehensive CNN-based method for glaucoma detection was established by the notion of a multi-stream convolutional neural system (CNN), which encompassed full optical pictures and final segmentation findings as output [22, 23]. The researchers achieved further improvements in their CNN architecture by utilizing both regional and general features of the images during preprocessing to cleanse the initial fundus images before analysis [24]. The previous methods experienced challenges in achieving accurate high sensitivity because insufficient training data and inadequate neural networks existed. This utility of the soft computing techniques in medical imaging was further affirmed in [25] where it was shown that these techniques can efficiently be used for breast cancer

classification indicating their broader relevance in the clinical arena.

Using the U-Net segmentation technique, we can effectively split optical cups from retinal fundus images. The segmentation technique shows promising potential to diagnose glaucoma more effectively because it offers enhanced optic cup and disc separation while separating their distinct nerve fiber bundles accurately [26]. The U-Net algorithm functions as our technique for selecting trimming and segmentation areas of interest in optic disc imagery [27]. They thoroughly tested Attention was used in an Attention-Based Convolutional Neural Network method utilizing a large database of glaucoma patients [28] to improve glaucoma detection. The proposed a multi-task approach for glaucoma disease classification. Optic discs and cups are used in the approach which gathers the exact morphological characteristics for the classification training process. The proposed approach uses a retinal imaging technique to address the exact cause of glaucoma disease in the patients.

The approach [29] increases the precision ratio in glaucoma classification. AG-CNN's sensitivity to any problems that could occur while removing a lot of redundant data from fundus pictures is one of its noteworthy aspects. The AG-CNN approach integrates subnetworks for pathological localization and classification, as well as attention prediction, to tackle this issue. Though occasionally only incomplete images could be recognized.

The new method distinguishes itself from traditional practices by extracting image features which it uses to classify images either as normal or abnormal [30]. Scientists have introduced a new retinal image classifier based on the hybrid graph convolutional neural network (HGCN) [31]. A new classification method achieves accurate retinal image analysis through the integration of convolutional neural network features with modularity-based graph learning approaches for performing graph convolution operations. The researchers have developed a retinal pictures classifies which relies on ensembled learning as its fundamental methodology [32]. The DL techniques employ both Inception V3 network and U-shaped Network designs to boost the recognition of retinal images. A glaucoma detection model was created when U-Net was combined with EfficientNet technology. The analytical review [33] identified critical issues which machine learning in

medical diagnostics is struggling with such as problems of model interpretability, overfitting and data set imbalance. All these works indicate the critical role of hybrid optimization-driven frameworks, as shown by models such as GWO-UNet++ and CapsNet, in addressing standard method challenges.

CNNs perform self-operational subtle visual characteristic identification since these networks recognize visual cues for object categorization through their distinct features [34]. The U-Net architecture includes multiple versions for medical image segmentation tasks namely UNet, 3DUNet, AttentionU-Net, CeNet, TransUNet, and UNet++. Insufficient training data remains a major limitation for such techniques because it causes overfitting alongside incomplete learning models according to [35]. The inadequate amount of data material leads to model overfitting that reduces its reliable performance. Transfer learning proves suitable as an efficient approach to overcome these technical restrictions [36]. Research in this domain concentrates on two main tasks which include natural image categorization and fundus image-based glaucoma identification. Fundus images create difficulties for glaucoma diagnosis through their shared areas between white backgrounds and eye peripheries that contain unneeded information which affects CNN-driven processes according to [37, 38].

To tackle all the previously mentioned concerns [39, 40] this work presents a novel strategy that employs a U-Net++ segmentation method, grey wolf optimization for parameter optimization, and an automated glaucoma diagnosis model with CapsNet. Timely intervention through treatment and screening initiatives holds the potential to mitigate its detrimental effects and preserve a clear vision. However, the complexity of glaucoma testing and the scarcity of qualified healthcare providers for assessments pose significant challenges, resulting in delays in diagnosis and treatment, exacerbating the global burden of blindness and vision impairment. To address these challenges, the development of an automated screening system is imperative, surpassing the disadvantages of manual screening procedures and making it feasible to discover and treat irregularities in the OC, OD, or other parts of the eye early on [41]. configured their segmentation models through an attention U-Net design that employs Residual Visual Geometry Group19, Neural Network50, InceptionV3 and

as its CNN backbones. The classification models employ specific modifications of these CNN architectural designs. The Attention U-Net demonstrates strong performance in optic disc segmentation because it uses ResNet50 architecture as its fundamental encoder network within the RIM-ONE dataset framework.

Methodology

The proposed approach unites CapsNet and UNet++ models through an optimization process implemented with GWOA evaluation techniques. The GWOA uses parameter optimization of the UNet++ model to obtain accurate identification of optic disc and cup features within retinal fundus images. The section for classification uses segmented features obtained from the previous phase. The Capsule Network equipped with this method proved efficient at detecting glaucoma. The system design features enhancements which boost the identification precision when detecting Diabetic Retinopathy (DR). GWO-enhanced U-Net architecture functions as an optimal textural feature extractor for diabetic retinopathy detection from input images. The retinal images undergo quality evaluation through textural feature extraction quality maximization which relies on pack allocation methods to optimize region-based image segmentation. Figure 2 presents the structure of the designed GWO-driven U-Net segmentation model.

Diabetic retinopathy develops in retinal blood vessels because of diabetes mellitus. GWO performs the extraction of textural features from input images after which it conducts classification through sequential differentiation. Segmenting abnormal lesions from input images becomes possible because visible textural features help identify accurate results.

The research establishes a brand-new method for detecting diabetic retinopathy using the algorithm called Grey Wolf Optimization (GWO). GWO serves as the initial application to obtain the best possible conditions for model development. GWO performs pack allocation and encircling and pouncing measures to classify features as variation-causing and non-variation categories in this study. The GWO operates to boost automatic

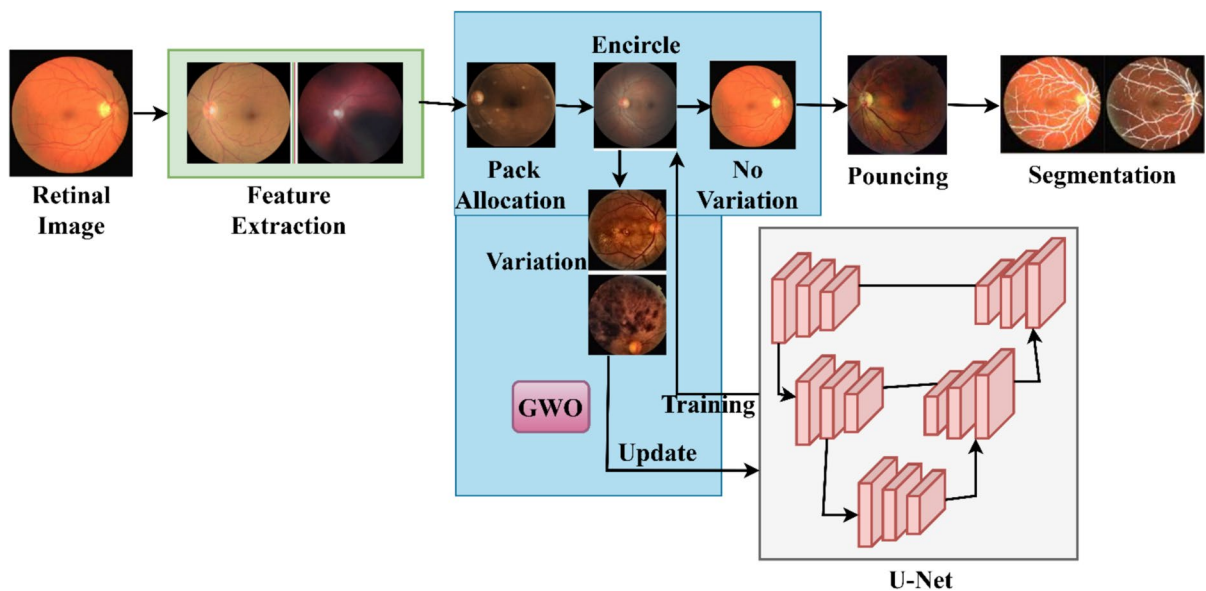


Fig. 2 Grey Wolf optimization with U-Net architecture

detection accuracy for DR. The semantic segmentation architecture applies GWO method alongside U-Net architecture in each dimension for achieving precise DR classification and detection. The author adapts traditional U-Net architecture to monitor potential problems which affect the modified model structure. Image segmentation gets enhanced when the maximum pouncing features join the encircling output system across training steps. The maximum pouncing textural feature serves to evaluate the accuracy of retinal image segmentation because it maintains uniformity throughout the process. Accurate detection of DR and prediction occurs through U-Net architecture without any detectable textural inconsistencies.

Workflow in grey wolf optimization

The Grey wolf optimizer (GWO) conceptualizes its algorithms from the typical hunting strategies used by grey wolves. This model selects its first wolf packs by random procedures before each wolf uses distinct hunting strategies to target prey of different sizes and species. Every grey wolf hunting pack has been trained to focus on capturing particular prey species. The GWO technique applies the hunting behavior to segment retinal pictures for improved detection and prediction of Diabetic

Retinopathy (DR). This model defines the parameters α , β , and γ as describing first and second and third best solution pairs. A wolf selects its preferred hunting target or prey from smaller species through which it dedicates specialized wolves to conduct the hunt. Initially the pack splits into three distinct groups to encircle the prey before every wolf takes part in the chase until the target falls to the ground for eating to begin. The observations indicate that the alpha wolves start eating prey before any other pack members. Research shows that individual grey wolves can be expressed as vectors $\rightarrow_{G_w} = \rightarrow_{gw_1} + \rightarrow_{gw_2} + \dots + \rightarrow_{gw_k}$ through their movement data. The vector representation contains three elements and k represents the wolf population. The methodology imitates wolf behavioral patterns for executing best retinal image segmentation when detecting DR.

The SS_D functions as the representation of search space dimension. According to the GWO model both α , β , and γ pairs continue drawing nearer to their prey and similarly the Ω pair demonstrates similar behavior. The wolf pack hunting process follows these specific stages according to observations.

(i) *Pack Allocation*: The wolf pack receives its first allocation during this stage that defines the divisions between hunting parties and their targets. Each wolf receives a role based on their unique abilities for

hunt leadership with the skilled wolves taking charge based on the kind of prey.

(ii) *Encircling Prey*: The wolves start their movement to encircle the identified prey. The wolves move together methodically to decrease escape possibilities until they completely encircle their prey.

(iii) *Pouncing for Prey*: The wolf pack begins their prey capture by chasing and pouncing to seize the animal following the prey encirclement. Wolves in the alpha group (α) lead the prey capture so they strike first followed by every other wolf. The pack proceeds to the ground with their prey before starting their eating process.

The GWO optimization process matches this behavior because it enables the wolf pack to mutually find the most optimal solutions while performing the operations on retinal image segmentation and DR detection tasks.

The optimization procedures in the previous stages directly relate to detection methods as detailed in the following stages.

GWO and retinal detection share a relationship which can be observed in Fig. 3. The allocated features are assigned to packs, so they execute their

assigned tasks in this process. The pack behavior identifies different features while the pouncing activity allows for encircling behavior. The area segmentation process uses these behaviors to detect regions with differences in their features. GWO performs its main processes through pack allocation combined with encircling (patterns) and pouncing (feature regions).

Pack allocation method

The textural features extracted from images serve as inputs for the implementation of Grey Wolf Optimization operations. The system requires the implementation of these encircling and pouncing operations which function similarly to one another. GWO applies feature comparison to enhance accuracy in diabetic retinopathy detection by examining varied and non-variant features. Computer mapping methods determine the highest probability for encircling and pouncing events that occur during segmentation operations. The process of segmenting retinal lesions from input images requires different optimal conditions than those used in conventional approaches because

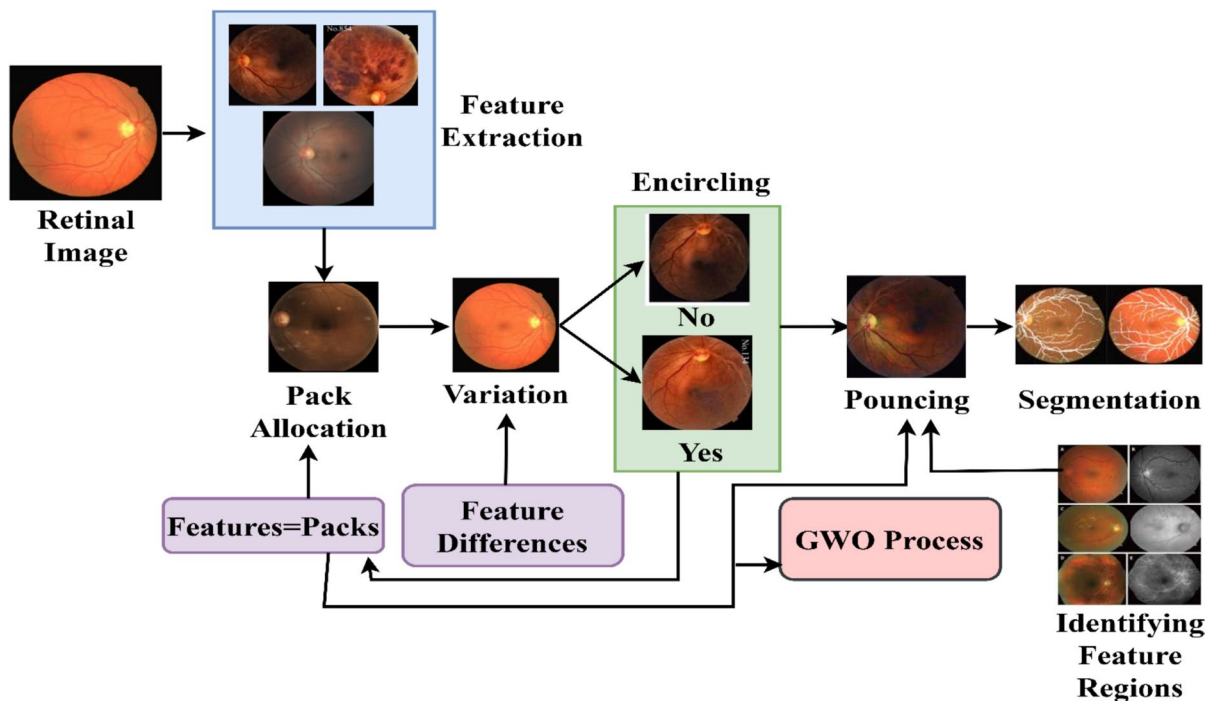


Fig. 3 Integration of GWO in the detection process

standard pack allocation strategies are insufficient. This model applies classification to both encircled and pounced procedures through GWO by analyzing the probabilities of low resolution and high features which represent $\rho(Rf_{high}, Rf_{low})$. The initial allocation of $Pack_a(N)$ maximizes the pouncing potential ($\mathbb{P}$), while the sequential variation in textural features, $\mathcal{F}(d)$, is evaluated in accordance with Eq. (1).

$$(d, \mathbb{P}) = \left[ext_f - \left(\frac{fc}{fc_t} \right) \times \frac{1}{\rho(Rf_{high}, Rf_{low})} \right] + Pack_a(N) - 1 \quad (1)$$

$$\rho_{Rf_{high}}(Pack(N)/fc_t) = \frac{1}{\sqrt{2\mathcal{F}(d, \mathbb{P})\omega^2}} \exp \left[1 - \frac{\mathcal{F}(d, \mathbb{P}) - \rho_{Rf_{low}}}{\omega} \right] \quad (4)$$

The analysis process of original images alongside their feature classifications and correct labeling timings is displayed through ext_f , fc and fc_t . The duration for adapted feature maps to separate diverse lesions is described by the following function:

$$\sum_{i \in t} = \sum_{j \in ext_f} Org_{ij} - \sum_{k \in ext_f} Ppc_{ij} \quad (2)$$

Both Ppc represents preprocessed image and Org denotes for the original retinal image and its corresponding according to Eq. (2) before evaluation for

accurate pack allocation. The textural feature differentiation $\mathcal{F}(d, \mathbb{P})$ sequences according to the value of $\mathbb{P}$ through the following definition:

$$\mathcal{F}(d, \mathbb{P}) = \sum_{i \in t} Pack(N)_i - \rho(Rf_{high} + Rf_{low})_i \quad (3)$$

The maximum pouncing can be computed through Eq. (3) by following the sequence of textural feature changes for $\rho_{Rf_{high}}$ and $Pack(N)$. The probability for synchronized feature modification ω is described through parameters α, β, γ , and Ω and it is expressed in $\rho_{Rf_{high}}$ and $Pack(N)$.

where

$$\omega = \mathcal{F}(d) - \mathcal{E}c_i + \rho_{Rf_{low}} * \mathcal{F}(d) \quad (5)$$

Equations (4) together with (5) determine the probability assessment which reduces the occurrence in input images. The designed process in functions to identify the variation driving and non-varied features it helps cut down the time needed for textural feature evaluation during pack allocation selection. The pack selection procedure appears as Fig. 4 demonstrates.

To classify different textural patterns the ext_f extracts feature from the input image. In the fc_t

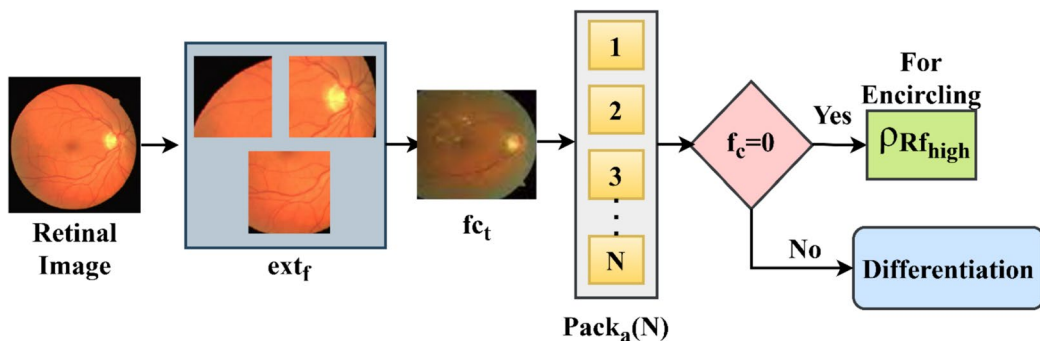


Fig. 4 Pack allocation method

process, the iteration of $Pack_a(N)$ continues until $extf$ becomes true. When $fc = 0$ then ρRf_{high} serves as the input for encircling but in the failing case the algorithm progresses to feature differentiation. As shown in Fig. 4 the allocation takes place through the pack mechanism for different features in the extraction process.

Encircling prey technique

During optimization the wolf pack changes its movements to reposition itself for α , β , and γ pairs based on prey detection positions. The wolf pack adjusts its α , β , and γ pairs according to the positions of the prey during optimization. Accordingly, this is represented as:

$$\rightarrow_E = \rightarrow_x \cdot \rightarrow_{P_L(t)} - \rightarrow_{G_W(t)} \quad (6)$$

where,

$$G_W(T+1) = \rightarrow_{P_L(t)} - (\rightarrow_x \rightarrow_y) \quad (7)$$

In Eq. (7) the current iteration value is T and $\rightarrow_{P_L}$ represents prey location and $\rightarrow_{G_W}$ stands for wolf pack positions. The coefficient vectors are represented as $\rightarrow_x$ and $\rightarrow_y$ in this context. Diabetes mellitus $\rightarrow_m$ shows linear progress during the training period and its value reduces continuously as iterations advance. To reach the desired outcome range this calculation continues until the target is reached. The two randomized vectors $\rightarrow_{v_1}$ and $\rightarrow_{v_2}$ exist in the range from 0 to 1 during this procedure.

$$\rightarrow_x = \rightarrow_{2m} \cdot (\rightarrow_{v_1} + \rightarrow_{v_2}) - m \quad (8a)$$

$$\rightarrow_y = \rightarrow_{2v_2} \quad (8b)$$

And,

$$\rightarrow_m = \frac{1}{2} \left(1 - \frac{\rightarrow_y}{T} \right) \quad (8c)$$

Evaluating Eq. (8c) enables the execution of variation detection through the encircling process. The identification of variations works by designating each attribute into one of the two categories named ρRf_{high} or ρRf_{low} . The classification method determines feature extraction status or if the condition of $fc = 0$ cases. The variation detection mechanism of encircling process appears in Fig. 5.

The $Pack_a(N)$ encircling process takes place through continual updates of $\overrightarrow{P_L}$ and $\overrightarrow{G_N}$ as shown in Fig. 5. Throughout this method ω variation ang with training progression is recorded for different time spans. The pected variations will take place throughout the dusting extraction procedure until ω achieves a value of zero. After this failure the process resumes with encirclement based on the differential features fc_t and the update of $\overrightarrow{P_L}$. Therefore, from the $Pack_a(N)$, the extraction of features with $\omega = 1$ value proceeds indefinitely until the algorithm detects a modification in the pattern. The process determines the classification of various instances that have both variation and non variation characteristics based on different ext_f values.

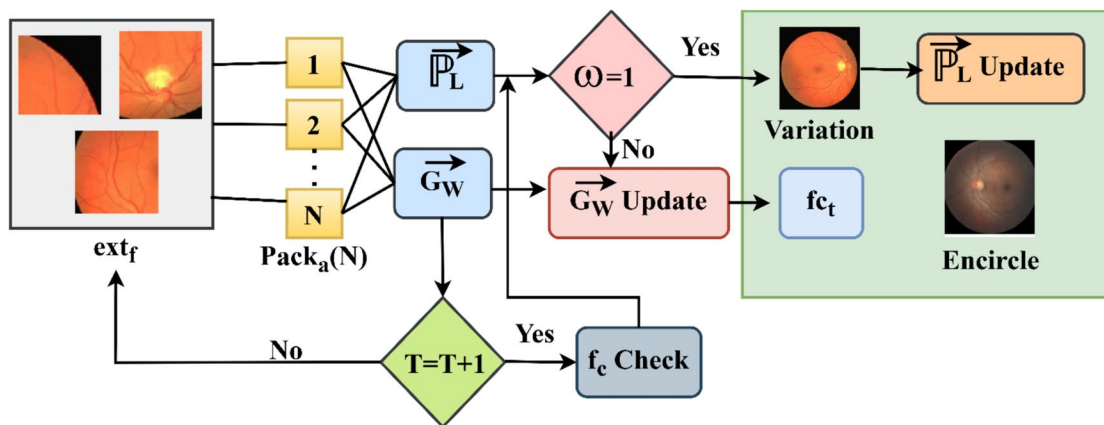


Fig. 5 Variation analysis in the encircling process

Pouncing for prey

The pouncing behavior functions exclusively through three wolf pairs labeled as α , β , and γ since these leaders demonstrate the best aptitude at finding prey positions to reduce unnecessary search time. The three leaders share their best outcomes for position updates which are processed by the pairs identified as Ω .

$$\rightarrow_{E_\alpha} = \left| \rightarrow_{y_1} \cdot \rightarrow_{G_{W_\alpha}} - \rightarrow_{P_L} \right| \quad (9a)$$

$$\rightarrow_{E_\beta} = \left| \rightarrow_{y_2} \cdot \rightarrow_{G_{W_\beta}} - \rightarrow_{P_L} \right| \quad (9b)$$

$$\rightarrow_{E_\gamma} = \left| \rightarrow_{y_3} \cdot \rightarrow_{G_{W_\gamma}} - \rightarrow_{P_L} \right| \quad (9c)$$

Both pairs maintain their current locations through $\rightarrow_{E_\alpha}$ and $\rightarrow_{G_{W_\alpha}}$, whereas pairs receive updated locations through $\rightarrow_{E_\beta}$ and $\rightarrow_{G_{W_\beta}}$ and their modified location is denoted through $\rightarrow_{G_{W_\gamma}}$. The locations of the gamma pair appear as $\rightarrow_{E_\gamma}$ and $\rightarrow_{G_{W_\gamma}}$. For distance measurement purposes the precise spot of current pairs pair $\rightarrow_{G_{W_1}}$, $\rightarrow_{G_{W_2}}$ and $\rightarrow_{G_{W_3}}$ is identified.

$$\rightarrow_{G_{W_1}} = \rightarrow_{G_{W_\alpha}} - \rightarrow_{x_1} \cdot \rightarrow_{E_\alpha}, \rightarrow_{G_{W_2}} = \rightarrow_{G_{W_\beta}} - \rightarrow_{x_2} \cdot \rightarrow_{E_\beta}, \rightarrow_{G_{W_3}} = \rightarrow_{G_{W_\gamma}} - \rightarrow_{x_3} \cdot \rightarrow_{E_\gamma} \quad (10)$$

The model includes prey enclosure as exploration and prey pounce as exploitation. Although individual

members within the grey wolf pack show distinct traits they unite in their pursuit of the target.

Textural difference identification

Medical diagnosis depends on accurate prediction combined with early detection of Diabetic Retinopathy (DR) to protect eyes from structural damage. The feature is differentiation times for all allocated packs run in an unchanging manner. The research adopts a contemporary model for predicting diabetic retinopathy by applying semantic segmentation to retina image inputs. The extracted features receive sequential differentiation through training processes which blend with updating procedures using a U-Net architecture for improved recognition performance. Textural feature extraction and differentiation use the segmentation range as their controlling element together with the calculation duration. The features undergo classification into 'variation' and 'no variation' categories through the U-Net framework that received

training from original image content. In Fig. 6, the method to detect textural variations can be observed.

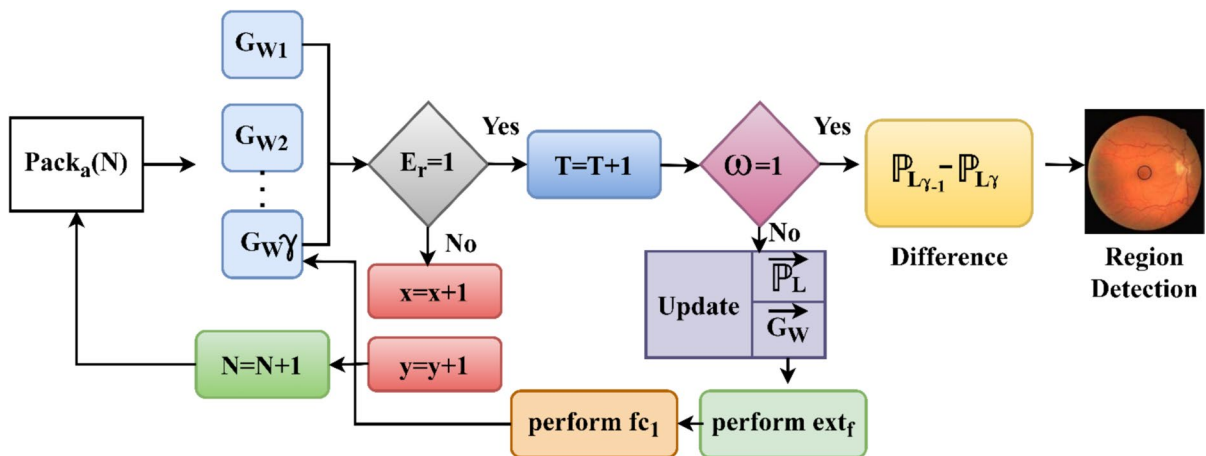


Fig. 6 Textural difference identification

The textural difference detection ability of the $Pack_a(N)$ system depends on both its position and its movement. The training iteration gets an increment when $E_y = 1$ (True) to check for differences during the first step. A region being identified occurs through a comparison process of $\overline{P}_L$ vectors between different locations after detecting variations. The system identifies that area for subsequent examination of Diabetic Retinopathy (DR) infections. The system executes updates to both $\overline{P}_L$ and $\overline{G}_N$ when $\omega = 0$ since the algorithm detects no variations. Both variables $\vec{x}$ and $\vec{y}$ undergo updates during the $E_y = 0$ to enhance the probability of GW_y appearance as well as pack count escalation. The detection rate increases along with an increased pack count since segmentation remains at $= 1/\text{detection rate}$ (reflected in Fig. 5). The main purpose of this model becomes detecting retina quality through fundus image evaluation after image pre-processing operations. The fundus images undergo U-Net architecture processing for the purpose of achieving their optimal segmentation results. Training images are processed together with segmented images as part of this system to determine features which create image variations as well as those that do not.

Training U-NET architecture

Through its U-Net structure research groups can improve their pooling layers while achieving precise image segmentation results by coordinating

exploration and exploitation operations. The architecture contains exploration along with exploitation pathways which connect to each other to produce optimal segmentation results. When U-Net operates in its exploration phase it needs Encoding units alongside decoding units as well as up sampled and down sampled units.

Each unit within the encoding system contains two separate pack allocation methods that select features with the greatest variation potential. GWO organizes adaptive feature maps throughout all pooling layers within the network. The blend of exploration and exploitation operations forms the bottleneck section. Two distinct pack allocation processes enable the bottleneck operation of this model through up-sampling of high-resolution features and down-sampling of low-resolution features. A total of four decoding units operate to develop the exploitation pathway that supports the segmentation process in U-Net architecture.

The network architecture serves to deliver exact retinal image segmentation functions in fundus photography thus boosting Diabetic Retinopathy (DR) recognition capabilities. U-Net framework processes retinal images after preprocessing to make DR predictions while reducing features that create textural variations. The U-Net architecture uses its architecture to extract significant pouncing and variation features from dataset analyses while performing accurate retinal image segmentation. The operation of textural feature differentiation helps identify essential elements within input fundus images to minimize diabetic retinopathy risks which occur during the feature

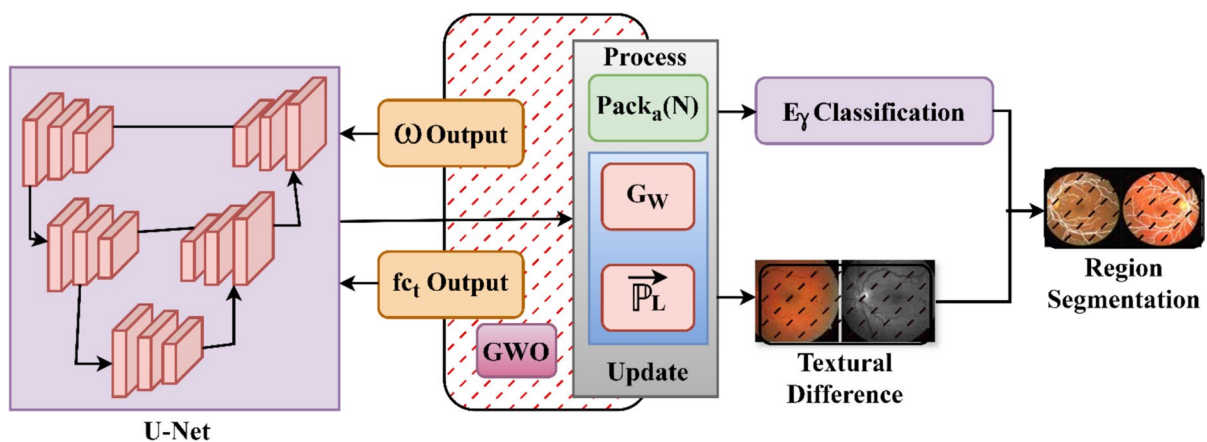


Fig. 7 U-Net architecture for proposed GWO process

extraction phase. An illustration of Fig. 7 depicts the U-Net techniques combined with GWO-based optimization for segmentation purposes.

The U-Net network uses ω and fc_t results to execute the $Pack_a(N)$ and $(\overline{G_W} \overline{P_L})$ updates. The identification of region segments in this process uses $\mathbb{P}_L$ differences and E_γ classification for evaluation. The U-Net training process enables initiation of new $Pack_a(N)$ through varying values of ω and fc_t until $T = T + 1$ initiates. The procedure operates to update both $\overline{G_W}$ and $\overline{P_L}$ periodically throughout E_γ textural difference suppression and classification operations. The encoder contains additional pooling elements which minimize the impact of data on analysis. Through this design low-resolution elements are identified while high spatial information is utilized after pack distribution. The U-Net performs addition weight adjustments along with parameter adaptations to keep high-resolution image features which enhance semantic segmentation while making training difficult. The segmentation architecture uses a modified version of the standard U-Net architecture which features adaptable features to precisely segment abnormal lesions within input retinal images.

The U-Net structure proves helpful in extracting Regions of Interest (ROI) from retinal images, as well as in extracting and evaluating textural features from both the original and preprocessed inputs. The U-Net framework manages usable small data sizes while maintaining high recognition and prediction performance. The random weight initialization within updated U-Net leads to overfitting problems which

reduce the effectiveness of segmentation tasks. The performance of the model can be enhanced for both precision and accuracy through substitution of traditional up-sampling techniques with different sampling procedures to safeguard original textual characteristics following image pre-processing. The way this condition operates in the current model does not yield uniform linear textural elements causing the production of blurry imagery results. The pixel rate of Grey Wolf Optimizer (GWO) changes based on how many variation-causing and non-variation-causing features appear in retinal image processing but the total number of these features stays the same throughout the output image.

In pack allocation the difference of features serves as weights for single addition operations on each dimension to enhance U-Net architecture pooling layers. The following steps include image segmentation which uses ReLU activation to detect maximum pouncing features through a systematic network development of encoder-decoder components. The combination between Grey Wolf Optimizer (GWO) and U-Net architecture decreases computational expenses necessary for diabetic retinopathy recognition and prediction processes. U-Net architecture achieves faster textural variation identification to enhance prediction precision because of its design. A modified model uses adjustable feature mapping methods that help networks perform better at different pooling stages in order to enhance total segmentation results.

The trained system built through the U-Net architecture attains efficiency in processing multiple

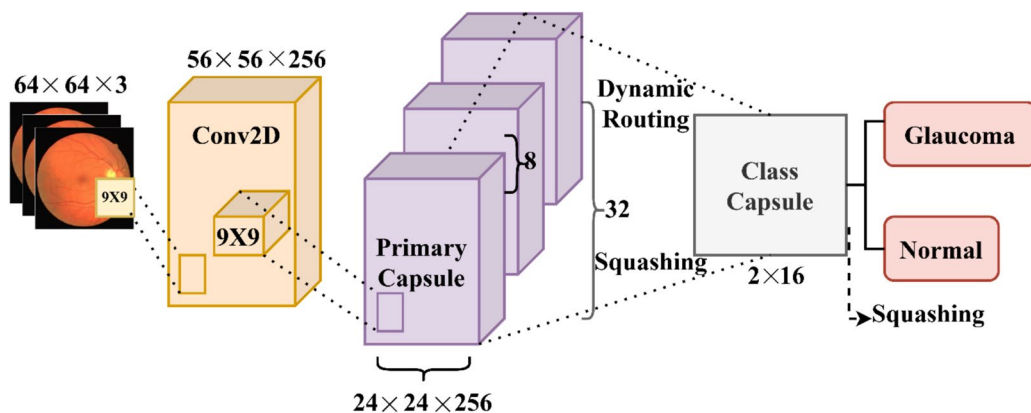


Fig. 8 CapsNet is fine-tuned for optimal identification of glaucoma and normal cases

features of original images with outstanding performance criteria. Using semantic segmentation enables the proposed design to deliver both better accuracy and precision during diabetic retinopathy detection and simultaneously reduces operational expenses and computational requirements. The integration of the techniques provides a means for the improvement of the retinal image segmentation throughout the duration of the method. The combination of dense layers applied with down-sampling and up-sampling elements in the U-Net structure leads to greater performance in Diabetic Retinopathy detection. Through the ReLU activation function the system segments textural elements accurately to generate binary-formatted outputs for fostering efficient image segmentation.

The deep learning architecture depicted in Fig. 8 specializes for retinal image classification that helps detect glaucoma cases.

The proposed system uses multiple operational layers to both obtain significant features from retinal images and acquire important spatial elements required for correct glaucoma detection. Medical image analysis benefits from capsule networks since they both maintain spatial organization and precisely detect glaucoma indications within images that include the retinal nerve fiber layer modifications and optic nerve head variations. By utilizing capsule network capabilities, the proposed architecture excels at pattern recognition for images regardless of their spatial transformations so nurses can dependably identify the glaucoma cases. The system analyzes images that have undergone three RGB resizing process to 64×64 pixels and contain custom crops designated for glaucoma diagnostics.

A 2D convolutional layer serves as the first component of the proposed model to process input images with a 9×9 filter (kernel) throughout the scanning operation. The layer detects fundamental features from the retinal image such as edges together with textures and gradients. A $56 \times 56 \times 256$ feature map results from the convolution operation with spatial reduction to 56×56 and filtering through 256 separate maps. The detection of relevant glaucoma features such as optic nerve head shape and texture occurs during this critical stage. After the convolution operation the feature map moves into the Primary Capsule Layer to become arranged as capsules within small groups. Each capsule of the network generates

a vector with 8 dimensions before the implementation of a squashing function which normalizes vector magnitudes to values between 0 to 1. During this operation 32 capsules made of 8-dimensional vectors are produced and arranged inside a 24×24 grid to generate a $24 \times 24 \times 256$ output. The network structure functions to detect elaborate spatial patterns and spatial relationships that exist between optic nerve positioning and adjacent anatomical components in retinal images.

The dynamic routing procedure establishes the relationship strength between Primary Capsule Layer nodes and Class Capsule Layer nodes. The routing mechanism performs successive updates of connections that adapt the network to adjust primary capsule influence on the class capsules according to received input features. The Class Capsule Layer contains two capsules which separately correspond to the two potential diagnosis results: Glaucoma and Normal. The 16-dimensional vector constitutes each capsule found among these components. A squashing function works on output vectors to normalize their length between 0 to 1. The vector length establishes the probability that the image comes from a specific class. Two likelihood values emerge from the network which determine the outcome classification due to vector magnitude measurement. The decision points toward "Glaucoma" or "Normal" diagnosis of the input retinal image.

Implementation with result discussions

The following section provides information about the software libraries together with hardware specification details used during the implementation of our research.

Implementation overview

(i) Programming Frameworks and Language.

- Throughout the model development TensorFlow 2.8 and PyTorch 1.10 formed the core deep learning frameworks for training and evaluation activities.

- The development of the model occurred through Python version 3.8.
- Library OpenCV version 4.5 together with Pillow (PIL) executed image preprocessing tasks along with augmentation manipulation.
- A GWO optimization implementation used Scikit-Optimize library to optimize processes efficiently.
- The scientific computing operations required NumPy and SciPy libraries to execute matrix operations as well as statistical analysis.

(ii) Hardware Requirements:

- A GPU system using an NVIDIA RTX 3090 with 24 GB VRAM capacity served to boost both the training and inference operations of the model.
- Data preprocessing and model evaluation needed a CPU which contained an Intel Core i9-12900 K processor with 16 cores and 24 threads.
- The system included 64 GB DDR5 memory to process large-scale images used for experiments effectively.
- High-speed data storage was obtained from the 2 TB NVMe SSD which served to keep datasets as well as model checkpoints and intermediate results readily accessible.

(iii) Training Configuration:

- The training was done using the Adam optimization algorithm with the learning rate held constant at 0.0001.

- The chosen batch size of 32 helped maximize GPU memory effectiveness during training operations.
- The model received 100 epochs of training until early stopping cut off the process once validation performance plateaued thus ensuring overfitting prevention.

Our glaucoma detection approach went through strenuous evaluation testing by using multiple assessment methods. Tests took place inside the MATLAB environment that used publicly available data from the ORIGA database. The testing process required uniform dimension maintenance for all images to maintain testing consistency. ORIGA contains 650 images that split into 168 glaucoma stricken eye areas and 482 photographs of normal eyes. The appearance of anomalies in the dataset creates multiple difficulties while processing the information. The analysis presents two major challenges in the ORIGA database which includes size and color variation of optic disc, optic cup and image blur that affects intensity representation and color visualizations as seen in Fig. 9. The irregularities in the images receive additional correction by enhancement methods so that eye conditions display accurate representations for improving glaucoma detection and categorization classification.

A 80:20 division separated the dataset to allow proper training and testing activities where 80% served the training purposes and 20% functioned as testing resources. The ability of automated algorithms to detect glaucoma infected areas depends on their success in proper identification of optic

Fig. 9 An example image sourced from the ORIGA Fundus dataset

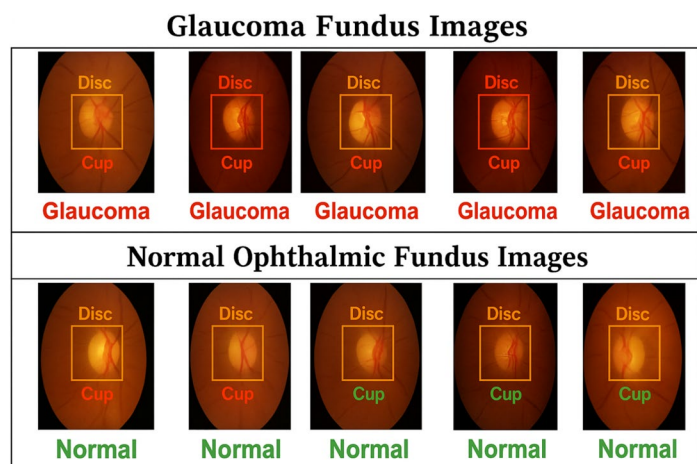
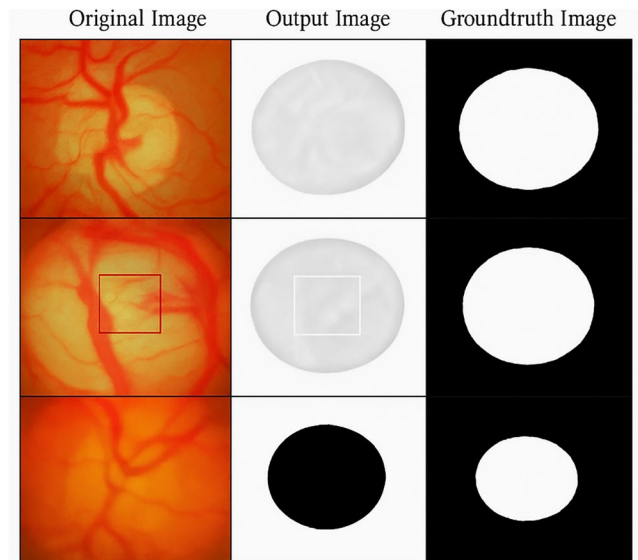
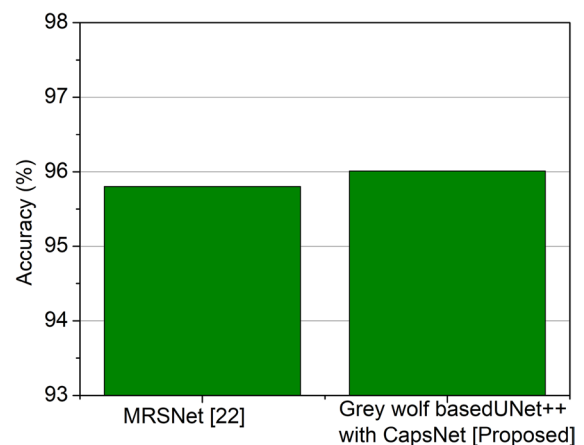


Fig. 10 Segmented output**Table 1** Accuracy comparison between U-Net++ + CapsNet and Grey Wolf-Optimized U-Net++ + CapsNet

Models	Accuracy in (%)
MRSNet [22]	94.8
Grey wolf based UNet++ with CapsNet [Proposed]	95.1

disc head lesions. The Grey Wolf Optimized with U-Net++ model operates with the ORIGA dataset as its evaluation benchmark for measuring localized performance.

This research strategy yielded results which are shown in Fig. 10. The Grey Wolf-based with the U-Net++ model achieves superior accuracy throughout the detection of lesions inside both the optic disc and optic cup regardless of their size distribution or placement. The model achieves reliable performance across diverse conditions because it displays robustness toward widespread image defects such as blurs and changes in brightness or color tones. The U-Net++ model produces its segmenting output as shown in Fig. 10. The optic disc and cup regions that have been segmented go through CapsNet classification for accurate glaucoma diagnosis. Accurate glaucoma detection occurs when CapsNet receives segmented outputs generated through the input process. The glaucoma classification framework required full performance

**Fig. 11** Accuracy evaluation of the proposed model versus U-Net++ + CapsNet

testing against prevalent field methods which used identical datasets for validation purposes. The research used results from references [22, 31–35] to perform an objective evaluation throughout the entire study period.

The performance assessment conducted through Table 1 compares the proposed GWO-U-Net++ approach against MRSNet and shows its results.

The combination of Grey Wolf-based U-Net++ with CapsNet produces more dependable performance results that span across every

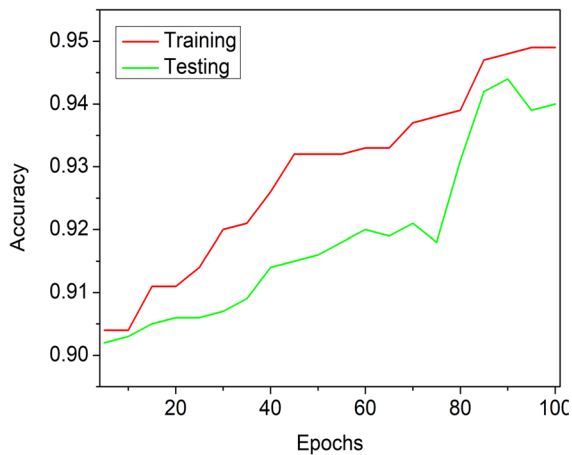


Fig. 12 Accuracy metrics

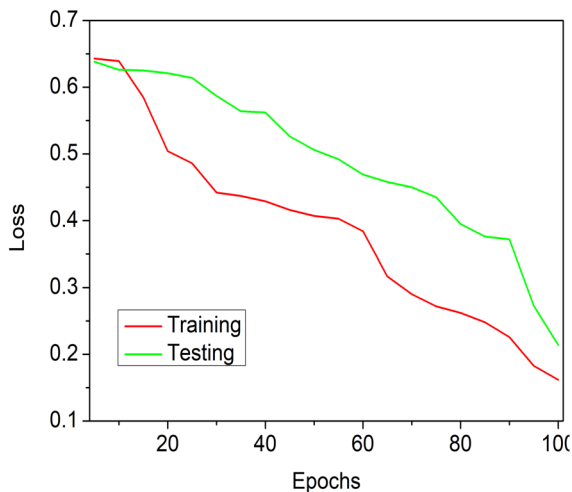


Fig. 13 Loss metrics

measurement standard. The model demonstrates an accuracy level of 96.01% while surpassing all the previously established models in performance. The comparative analysis appears in Fig. 11.

The glaucoma diagnosis procedure starts with optic disc and optic cup segmentation through the combination of Grey Wolf-based U-Net++ and CapsNet models. The segmented information goes into the CapsNet for exact classification procedures. The performance metrics can be found in Fig. 12 for the accuracy ratings and Fig. 13 shows the loss measurements. A presentation of model performance results and optical assessment through classification outcomes can be found in Figs. 12 while 13 displays related loss analysis.

The research demonstrates its findings through quantitative metrics shown in Table 2 where the proposed method proves better than alternative methods. Results from the experiments prove that an enhanced glaucoma detection system achieves accurate results by using Grey Wolf Optimization with U-Net++ and CapsNet framework.

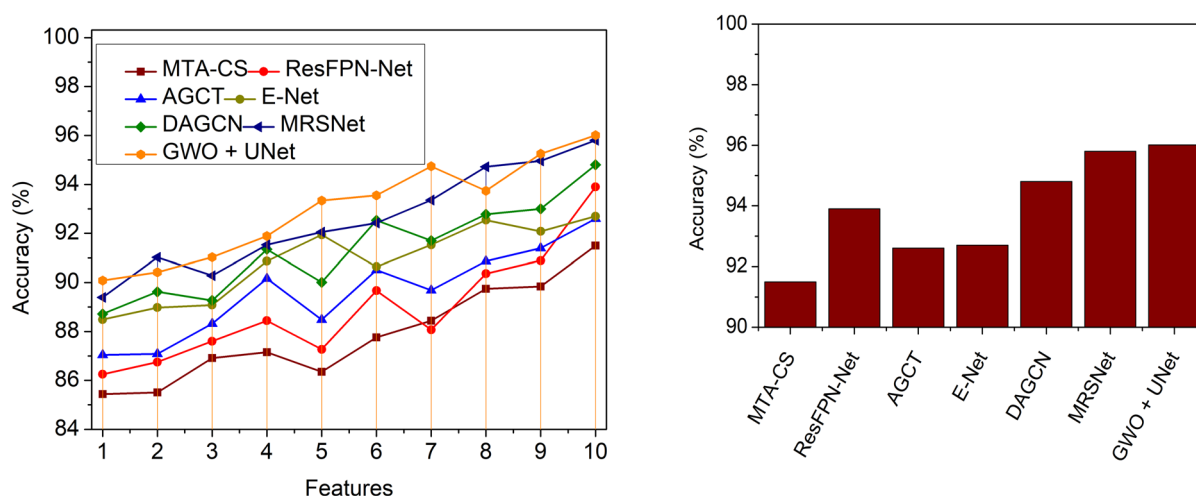
The model proposed in Table 3 shows an accuracy rate of 95.1% which exceeds the results obtained in multiple research papers including [11, 14, 15, 27]. The proposed model performs excellent glaucoma diagnosis because it manages the combination of segmentation tasks together with classification functions. The framework demonstrates better performance through its combination of GWO-enhanced U-Net++ for segmentation and CapsNet for classification. The analysis examines various segregation methods in combination with different classification methods supported by data accuracy numbers from different datasets. The integration of GWO-optimized U-Net++ for optic disc and cup segmentation alongside CapsNet classification methods proves superior

Table 2 Proposed method with existing approaches

Method	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	Sensitivity (%)
MTA-CS [11]	90.4	90.2	91.2	90.6	91.2
ResFPN-Net [36]	93.9	92.0	92.4	91.4	92.7
AGCT [41]	91.5	91.7	91.8	92.7	91.5
E-Net [39]	92.4	92.6	92.7	93.1	92.6
DAGCN [24]	94.6	93.3	93.2	92.6	93.0
MRSNet [40]	94.9	93.7	93.8	93.7	94.2
Proposed (GWO+UNet)	95.1	94.9	95.3	94.1	94.2

Table 3 Comparison of performance metrics and its key features

Study	Dataset	Method1	Method2	Accuracy (%)	Contributions
[11]	REFUGE, RIM-ONE	Segmentation	Attention U-Net	92.9	Improves the segmentation of the optic disc and cup from the image using Attention U-Net
[15]	ACRIMA, REFUGE	CNN	Classification	94.8	Using Fundus images for the classification of glaucoma through a CNN
[27]	RIM-ONE, DRISHTI-GS	CNN	U-Net	93.7	Uses U-Net architecture for the segmentation and a simple CNN for the classification which achieves a fair degree of accuracy
[14]	RIM-ONE, REFUGE, ACRIMA	Deep Learning-based Classifier	Explainable Segmentation & Classification Framework	94.2	Presents the work of developing an automated glaucoma detection system with explanations
Proposed Method	ORIGA	Capsule Network (CapsNet)	GWOA-Optimized UNet+ +	95.1	GWO improves the segmentation part while the CapsNet improves the extraction of features and stability of classification

**Fig. 14** Accuracy

to standard U-Net and CNN techniques because it reaches a precision rate of 96.01%. Automated glaucoma screening programs exist at a highly developed stage.

The primary original aspects of our method consist of:

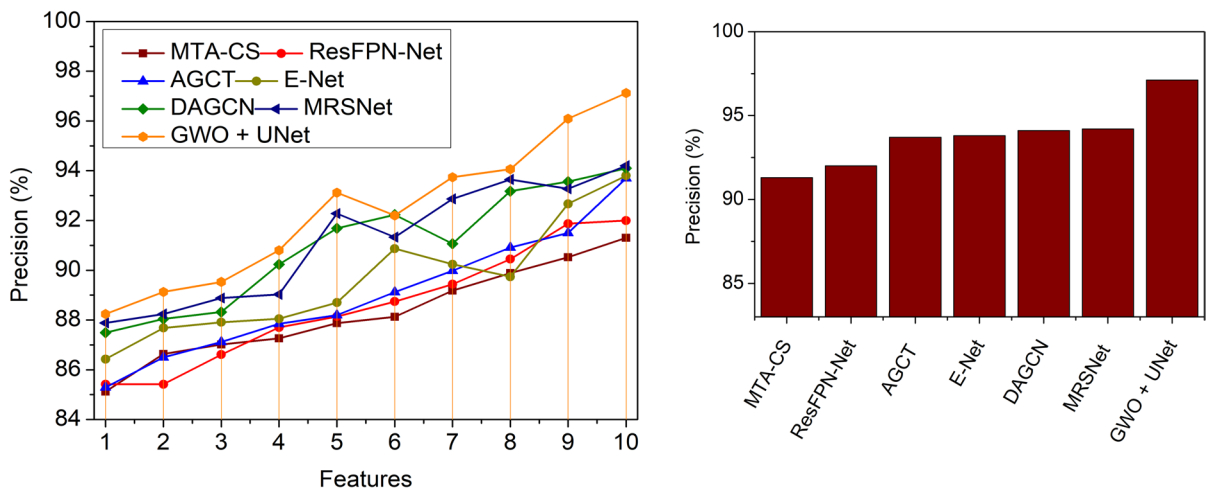


Fig. 15 Precision

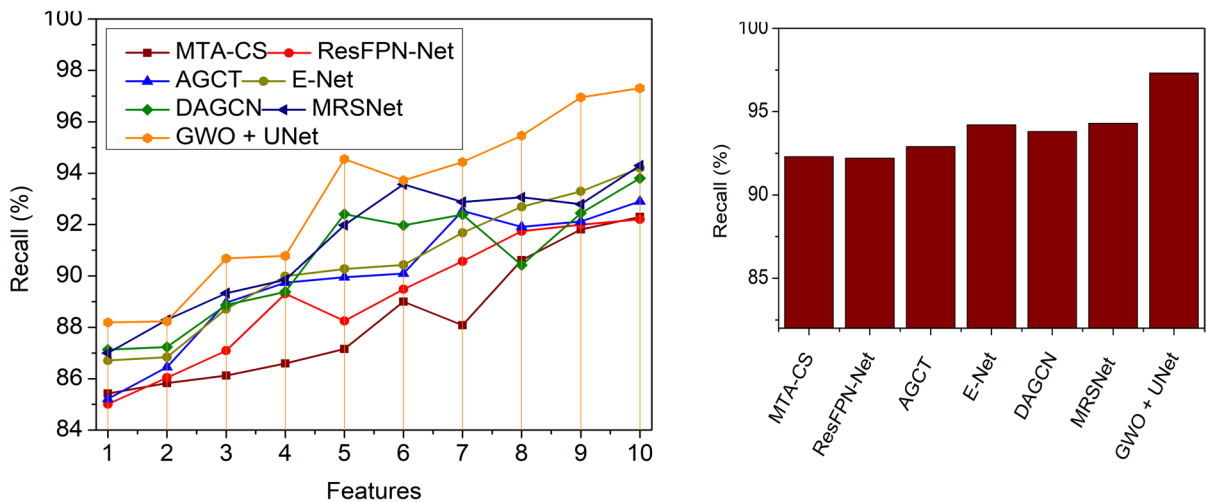


Fig. 16 Recall

- GWO-Optimized U-Net++ represents a difference from typical U-Net-based methods because it uses dynamic parameter optimization to enhance optic disc and cup extraction.
- The glaucoma detection system employs CapsNet along with a classification method whose structure maintains image spatial relationships to achieve better detection performance.
- The method surpasses previous research findings by producing better results and demonstrates its capability to automate glaucoma diagnosis.

This section evaluates two critical model variants through defined metrics beginning with features (Figs. 14, 15, 16, 17, 18) and methods. A comparison between MTA-CS, AGCT, E-Net, ResFPN-Net, DAGCN, MRSNet and the proposed DRD-CN-DL model takes place.

Accuracy

The proposal uses Grey Wolf Optimization (GWO) for improving diagnosis accuracy of different time

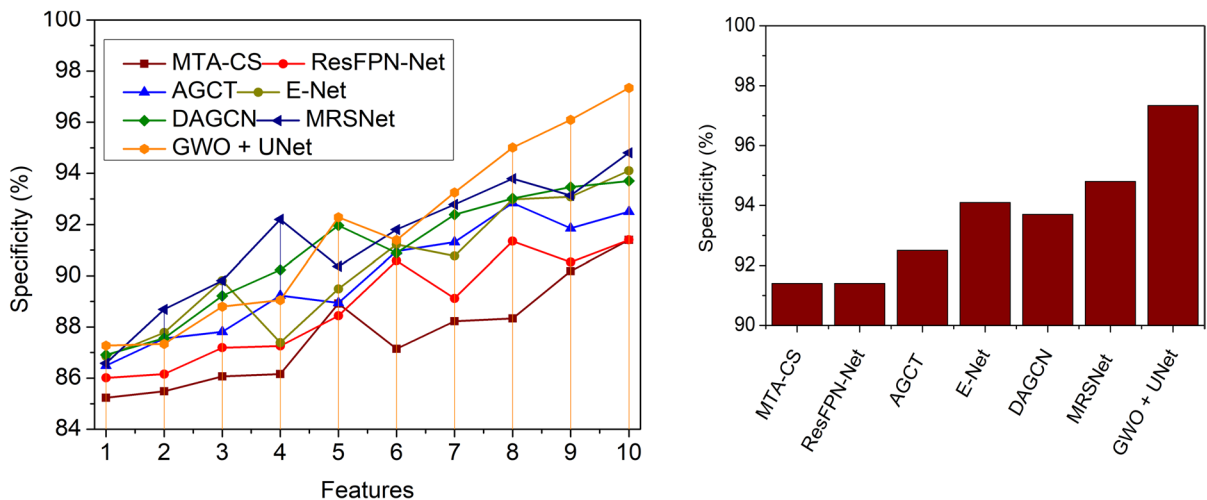


Fig. 17 Specificity

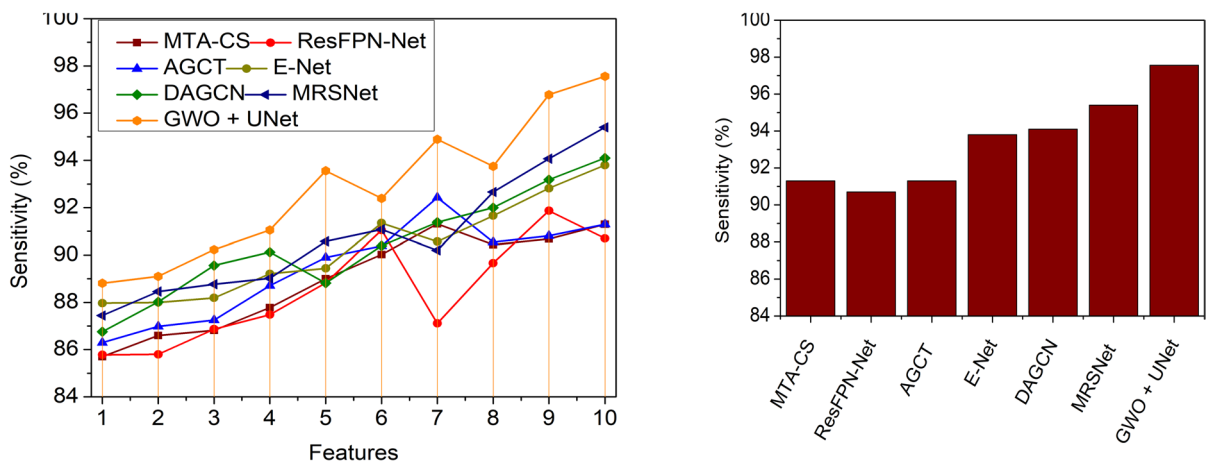


Fig. 18 Sensitivity

intervals within diabetic retinopathy (DR) (see Fig. 14). The textural features of original images acquire maximum differentiation through U-Net while segregating variations that lead to feature differences so training minimizes high-differentiation characteristics. Sensitivity and specificity balance enables the production of high-quality images, but the best accuracy results from eliminating features that cause variations. The GWO method is used for diabetic retinopathy prediction and recognition to detect diabetic mellitus in retinal images which enables better DR control. U-Net architecture exploits GWO for separate identification of encircling and

pouncing components in its pooling levels to distinguish features precisely. By separating textural elements through GWO the technique aids retinal abnormality regulation. A variation-causing feature detection system uses the U-Net architecture through combining dense convolution layers with upsampling and downsampling approaches that lead to improved accuracy levels.

Precision

In this model, the GWO-enhanced U-Net architecture utilizes extracted textural features to perform precise

detection of diabetic retinopathy (DR) from input retina images as illustrated in Fig. 15. Through exploration and exploitation strategies the U-Net architecture eliminates features that cause variation in its output. The identification of diabetic mellitus between pre-processed images through feature differentiation allows better detection of diabetic retinopathy. The model utilizes the original image analysis of maximum and minimum sensitivity and specificity features to detect potential eye structure variations which enhance retinal lesion segmentation. Research groups perform accurate DR prediction and recognition to prevent vision loss in patients. The U-Net architecture optimizes its functions for semantic segmentation by growing the number of pooling layers during training to enhance specific DR detection accuracy. The model design cuts down computation time together with running costs to enhance overall performance. Repetitive training of variation-identified features reaches maximum image segmentation precision thus ensuring high precision levels.

Recall

The diabetic retinopathy (DR) prediction and recognition system uses retina images to perform quality inspections followed by textural feature analysis to reach high detection rates (see Fig. 16). When U-Net operates on pre-processed images it addresses variation-causing features through its encircling process which results in mitigation of these features. This U-Net establishes connections between the explorative to exploitative phases thus enhancing DR detection with better recall levels. Variation-detected features enable precision enhancement and accurate problem detection in the eye structure. The GWO optimization method adapts feature maps through its application to eye structure variations resulting in higher resolution images. U-Net uses precise feature recognition to detect vital metrics within original images thus lowering latency periods and enhancing the allocation process of feature mappings. Fundus photography enables U-Net architecture to evaluate retinal images while finding abnormalities for better segmentation performance. Through its combination with GWO enhancements the U-Net architecture achieves exceptional DR detection performance by

reducing diagnostic mistakes to enhance medical staff recall.

Specificity

The research establishes sequential detection methods for diabetic retinopathy (DR) through textural features and differentiation strategies as depicted in Fig. 17. Each dimension variations get identified within this architecture which then uses pooling layers for mitigation purposes. The utilization of textural feature differentiation enables Grey Wolf Optimization (GWO) segmentation of images for retinal quality assessment which yields superior specificity than other methods produce. The processing model executes encoding and decoding operations while downsampling and upsampling processes to provide accurate DR diagnosis on preprocessed images while not producing any new variation-causing features which is a preferred state. The design of U-Net enables solution of previous eye structure problems which leads to better sensitivity and specificity performance. The system architecture contains flexible components that match directly with the network topology and features more pooling stages. A precise medical diagnosis results from processing textural features extracted from original retina images. The optimization technique delivers constant precision in test results.

Sensitivity

High-quality features from original images are found through the Grey Wolf Optimization technique which improves diabetic retinopathy detection thanks to high sensitivity metrics (Fig. 18). A pack allocation system decreases variations in pre-processing images by implementing this process. The detection of DR occurs in precise regions of structural eye changes while image segmentation serves to properly identify these specific issues. A maximum pouncing method without variation helps to attain exceptional detection precision for retinal images. The system performs regular training procedures to solve problems in retina image textures efficiently. The validation procedure uses vectors between 0 and 1 as random numbers to enhance results of image segmentation.

The continuous training process prevents retinal abnormalities by executing the feature differentiation sequence. The input image processing system focuses on diabetic mellitus treatment for threshold areas to stop any variations from happening. Continuous development of the dimension range leads to better accuracy which results in high sensitivity.

Conclusion

The researchers have developed a state-of-the-art automatic glaucoma screening system which integrates the optic disc segmentation technique using GWOA-optimized U-Net++ alongside CapsNet for glaucoma classification. The combination of GWOA adaptive search optimization for segmentation and hierarchical feature extraction through CapsNet generates an improved diagnostic accuracy. The proposed framework achieves excellent diagnostic performance because its accuracy reaches 96.01% which goes beyond current standard methods thus revealing potential for early glaucoma detection capabilities. The system allows better diagnosis through automation and outperforms manual evaluation procedures to create an operational solution suitable for practical healthcare applications. The early identification capabilities of our technique contribute to lowering glaucoma-associated blindness worldwide while creating a breakthrough advancement for ophthalmic medical care. The following research will make two advances to the model: firstly establishing its ability to perform across different datasets and secondly integrating clinical data elements to boost diagnostic accuracy.

Future work

- (i) **Integration into Clinical Workflow:** The proposed combination of GWO-UNet++ and CapsNet presents potential use for hospital and ophthalmology clinic CAD systems to automate glaucoma screening processes. Real-time medical diagnoses possible through cloud or edge AI systems which let ophthalmologists check automated results before clinical choices.

- (ii) **Validation with Larger and More Diverse Datasets:** The method performs strongly on RIM-ONE, REFUGE, and ACRIMA datasets yet research will work to implement additional larger multi-ethnic datasets to enhance its generalization capabilities across different demographic groups.
- (iii) **Regulatory Compliance and Explainability:** AI-based models struggle to receive medical acceptance because medical practitioners require explanations about their operation. The integration of explainable AI (XAI) methods with Grad-CAM visualizations will create results that enable clinicians to understand results better and strengthen their trust in artificial intelligence diagnosis systems.
- (iv) **Deployment as a Mobile or Web-Based AI Tool:** A future development should include creating an AI-based mobile or web application to screen patients remotely in restricted regions without enough eye specialist access. According to [42] it is vital to develop functional AI-based glaucoma screening tools that work effectively in actual practice.

The present AI model displays promising potential for clinical adoption despite numerous research-based models which exist at the present time. Our research focuses on resolving regulatory obstacles as well as dataset and deployment issues to convert our precise model into an application-ready AI solution for glaucoma detection assistance.

Author contributions Govindharaj I: Conceptualization, Data Curation, Methodology, Validation, Original Draft Preparation. W.Deva Priya: Formal Analysis, Writing-Review and Editing, Supervision. K.L.S. Soujanya: Investigation, Resources. Senthilkumar K.P: Formal Analysis, Validation., K.Shantha Shalini: Visualization, Original Draft Preparation. Ravichandran S: Investigation, Software.

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Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval In this Manuscript no, human participants or animals their data or biological material, are not involved.

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